Lab 08 -Sequences and Summations

Objective

Solving exercises from the text book in chapter 2.4

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to:

1. will understand Sequences and summations
2. They will be able to solve shorter/easier or longer / harder problems given in the textbook.

Lab Requirements

Students allowed using their lecture notes in the lab and use blackboard slides in order to solve the exercises.

Lab Assessment

1- Divide students to groups and let them to solve the given example.
2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

In this lab, the following exercises are going to be solved and explained to them:

1. What are the terms $a0$, $a1$, $a2$, and $a3$ of the sequence $\{an\}$, where an equals
   a) $2^n + 1$? **b)** $(n+1)^{n+1}$?
   c) $\lfloor n/2 \rfloor$? **d)** $\lfloor n/2 \rfloor + \lceil n/2 \rceil$?
2. List the first 10 terms of each of these sequences.
   a) The sequence that begins with 2 and in which each successive term is 3 more than the preceding term
   b) The sequence that lists each positive integer three times, in increasing order
3. Find the first five terms of the sequence defined by each of these recurrence relations and initial conditions. Then find their formulas.
   a) $a_n = 6a_{n-1}, a_0 = 2$
   b) $a_n = a_{n-1}^2, a_1 = 2$
4. For each of these lists of integers, provide a simple formula or rule that generates the terms of an integer sequence that begins with the given list. Assuming that your formula or rule is correct, determine the next three terms of the sequence.
   a) 3, 6, 12, 24, 48, 96, 192, . . .
   b) 15, 8, 1,−6,−13,−20,−27, . . .
5. Find the n^{th} term $\{a_n\}$ of the following:
   a) The geometric sequence with a_0= 4 and a_4=¼.
   b) The arithmetic progression with a_1=2 and a_4= -7.

6. What are the values of these sums?

a) $\sum_{k=1}^{5}(k+1)$ b) $\sum_{j=0}^{4}(-2)^{j}$

c) $\sum_{i=1}^{10} 3$ d) $\sum_{j=0}^{8}(2^{j+1}-2^{j})$

7. What is the value of each of these sums of terms of a geometric progression?

a) $\sum_{j=0}^{8} 3 \cdot 2^{j}$ b) $\sum_{j=1}^{8} 2^{j}$

c) $\sum_{j=2}^{8}(-3)^{j}$ d) $\sum_{j=0}^{8} 2 \cdot (-3)^{j}$

8. Compute each of these double sums.

a) $\sum_{i=1}^{2}\sum_{j=1}^{3}(i+j)$ b) $\sum_{i=0}^{2}\sum_{j=0}^{3}(2i+3j)$

c) $\sum_{i=1}^{3}\sum_{j=0}^{2} i$ d) $\sum_{i=0}^{2}\sum_{j=1}^{3} ij$

9. Find $\sum_{k=100}^{200} k$

10. Find $\sum_{i=1}^{3}\sum_{j=1}^{i} ij$